AI-Assisted Mixed-Signal Physical Design Using OpenLane and SKY130

Rishanya Shamganth

***Abstract*—This report presents the reproduction of a mixed-signal physical design flow using OpenLane and the SKY130 PDK. The design integrates the analog macro AMUX2_3V (2:1 analog multiplexer) with digital logic (design_mux). The complete RTL-to-GDSII flow was executed successfully. AI tools were used to understand the repository, debug issues, and organize documentation. The final results show DRC=0, LVS clean, and timing met.**

***Index Terms*—Mixed-signal design, OpenLane, SKY130, Analog macro, Physical design, AI-assisted workflow.**

I. Objective

The objective of this task was to study the fundamentals of mixed-signal physical design and reproduce a similar RTL-to-GDSII flow using open-source tools. The work includes repo study, analog macro integration, OpenLane configuration, synthesis, placement, routing, signoff verification, and layout visualization.

II. Reference Repository Summary

The reference repository (**vsdmixedsignalflow**) demonstrates the integration of an analog multiplexer macro (AMUX2_3V) into a digital design using OpenLane. The macro is abstracted using LEF, LIB, and Verilog blackbox views. LEF provides physical information (size, pins, routing layers), LIB provides timing/power abstraction, and the Verilog model preserves the macro during synthesis, OpenLane then performs synthesis, floorplanning, placement, PDN generation, routing, DRC, and final GDSII generation.

III. AI-Assisted Workflow

AI tools (ChatGPT) were used throughout the project for:

- Understanding mixed-signal concepts
- Explaining LEF/LIB/DEF/GDS formats
- Breaking the flow into milestones
- Analyzing synthesis and implementation errors
- Debugging OpenLane configuration issues
- Interpreting timing and signoff reports
- Organizing documentation and report writing

All AI-generated suggestions were manually verified through terminal outputs, reports, and layout views.

IV. Experiments and Debugging

The AMUX2_3V macro was inspected in Magic to verify pin placement, routing layers, and physical dimensions. The design_mux top module integrates this macro with digital logic. During the flow, several issues were encountered and resolved:

- Duplicate spi_slave module definitions
- Signal naming mismatch (IO vs I0)
- Analog macro declaration errors
- Synthesis failures in Yosys
- OpenLane configuration errors
- Magic visualization and DEF/LEF related issues

After iterative debugging, the flow completed successfully.

V. Results

The OpenLane flow completed successfully and generated all expected outputs.

Metric	Value
Flow Status	Completed Successfully
Synthesized Cells	301
Total Cells	15256
Critical Path	6.56 ns
Clock Period	10 ns
Suggested Frequency	100 MHz
DRC Violations	0
LVS Status	Clean
Setup Timing	Met
Hold Timing	Met

Key signoff results are shown in Fig. 1 and Fig. 2.

```
[INFO]:  Design                  : design_mux
         TECHNOLOGY LIBRARY      : sky130A
         SYNTHESIS STRATEGY      : AREA 1
         CIRCUIT DESIGN          : design_mux
         VERILOG FILE            : merged.v
         SYNTHESIS REPORT        : reports/synthesis
[SUCCESS]:  Flow complete.
-------------------------------------------------
Elapsed time: 0h3m39s
-------------------------------------------------
```

Fig. 1. OpenLane flow completion message.

```
  Total      Magic     Netgen     Netgen     PEX     CALIBRE
   DRC        DRC       LVS        LPE       PEX       xRC/
Violation   (short)    (Com)      (LVS)              (vs GDS)
---------------------------------------------------------------
    0          0         0          0         -         -

Total DRC violations       =  0
Total LVS errors           =  0
Total warnings             =  0
Total PEX errors           =  0
Total xRC errors           =  0

Design is LVS clean
```

Fig. 2. Manufacturability report (DRC & LVS results).

The final layout viewed in Magic is shown in Fig. 3.

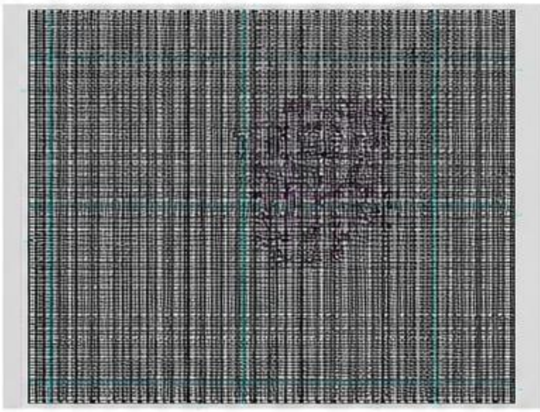

Fig. 3. Final GDS layout of design_mux viewed in Magic.

VI. Observations

The project demonstrated how analog macros can be successfully integrated into a digital implementation flow using LEF abstractions and OpenLane. Proper abstraction, signal consistency, and verification at each stage were critical for a clean signoff.

AI assistance significantly reduced the time required to learn EDA tools, debug issues, and document the workflow.

VII. Conclusion and Future Work

A complete mixed-signal RTL-to-GDSII flow was successfully reproduced using OpenLane and the SKY130 PDK. The design passed DRC and LVS checks and met timing requirements.

Future work includes multi-macro integration, timing/power optimization, and scaling the design to larger mixed-signal subsystems.

References

[1] P. M., "vsdmixedsignalflow," GitHub Repository, 2023. [Online]. Available: https://github.com/praharshapm/vsdmixedsignalflow

[2] The OpenLane Project, "OpenLane Documentation." [Online]. Available: https://openlane.readthedocs.io

[3] SkyWater Technology and Google, "SKY130 Open Source PDK." [Online]. Available: https://github.com/google/skywater-pdk

[4] The OpenROAD Project, "OpenROAD – Open Source Physical Design Tools." [Online]. Available: https://openroad.readthedocs.io